Pranshu Panda

work.pranshupanda@gmail.com | github.com/pranshupanda | linkedin.com/in/pranshu-panda

EDUCATION

M.S. Computer Science, University of Minnesota - Twin Cities **Expected May 2028**
Relevant Coursework: Advanced Machine Learning, Cloud Computing

B.S. Computer Science, University of Minnesota – Twin Cities **May 2026**
GPA: 3.74/4.00 | 5x Dean's List
Relevant Coursework: ML, NLP, Robotics, Numerical Algorithms, OS, Software Development, UI, Algorithms and Data Structures

SKILLS

Languages: Python, C, SQL, Java, Bash, HTML/CSS, TypeScript
AI/ML: Deep Learning **(**Transformers, CNNs), LLMs, Fine-Tuning, Bayesian Optimization, Hyperparameter Optimization, Reinforcement Learning, Supervised/ Unsupervised Learning, Search Algorithms, GPU Acceleration
Libraries/Frameworks/Tools: PyTorch, BoTorch, Scikit-learn, Pandas, OpenCV, Git, Hugging Face, ROS2

WORK EXPERIENCE

Graduate Research Assistant, University of Minnesota | Minneapolis, MN **Sept 2026 - Present**
- Leveraging LLM agents with evolutionary algorithms to automate experimental design for protein engineering.

Undergraduate Teaching Assistant, University of Minnesota | Minneapolis, MN **Sept 2025 - May 2026**
- Led weekly labs and debugging sessions for 600+ students in advanced programming principles and computational linear algebra; mentored students on recursion, software testing, functional programming concepts, and algorithmic problem solving.

Undergraduate Researcher, University of Minnesota | Minneapolis, MN | GitHub **March 2025 - May 2026**
- Researched probabilistic ML for experimental design applied to molecules and proteins under Prof. Aryan Deshwal.
- Built GPU-accelerated bayesian optimization pipelines in PyTorch for 60-dimensional combinatorial optimization tasks.
- Implemented evolutionary (genetic) search strategies, achieving a 3.6% performance improvement over standard PyTorch baselines.

Research Intern, Indian Institute of Technology Delhi (CSE) | Delhi, India **June 2024 - Sept 2024, Dec 2025 - Jan 2026**
- Developed a PyTorch-based hyperparameter optimization framework to automate throughput tuning for cache-management policies in a black-box systems environment.
- Built a Python visualization and debugging tool to render queue states, packet flows, and request latencies from simulation traces.

Software Engineering Intern, Stryker | Delhi, India **June 2023 - Aug 2023**
- Fine-tuned a YOLOv7 object detection model on a custom 3,000-image dataset, achieving 0.88 Mean Average Precision (mAP) for real-time operating-room equipment detection for Endoscopy R&D.
- Refactored UI layout and implemented cross-device responsive design for the OCULAN Surgical Light service tool across multiple clinical screen resolutions.
- Worked with the Sustainability Solutions team on software and tooling to support reprocessing of single-use medical devices.

RESEARCH AND PROJECTS

TOST: Threshold-based Object Sorting Technique | GitHub **Jan 2026 - May 2026**
- Developed a ROS2-based autonomous real-time object sorting system in Gazebo, utilizing an OpenCV color-segmentation pipeline for centroid localization to drive a 5-DOF robotic arm to precisely and efficiently sort blocks into designated bins.

Fine-Tuning of Transformers with LoRA: A Comparative Study | GitHub **Nov 2025 - Dec 2025**
- Benchmarked Low-Rank Adaptation (LoRA) against full-model and head-only fine-tuning on SST-2 and IMDB; demonstrated that LoRA achieves superior accuracy on SST-2 (94.27% vs 93.81%) and parity on IMDB (~95.5%), validating PEFT as a robust alternative to computationally expensive full fine-tuning.

Shortest Paths for Trivially Partitioned Graphs | GitHub **Sept 2024 - Dec 2024**
- Optimized and analyzed Dijkstra's Algorithm and A* Search in Python using a hierarchical graph partitioning strategy, achieved up to a 52.7% runtime reduction on large trivially partitioned graphs.

WINGS - University of Minnesota Rocket Team **Sept 2023 - Dec 2024**
- Collaborated on the third-gen ground station (WINGS) with the Avionics team using TypeScript/React/UnoCSS for live rocket telemetry; created a custom simulator to test serial I/O and validate end-to-end data flow through Rust API pipelines.